APPENDIX

Survey Sample Dataset

	A	B	C	D	E	F	G	H	I	J
1	Timestamp	Your Name	Age Group	Gender	How often do you shop online?	Which shopping app do you use the most	Have you ever added a product to your sa	Did you purchase the product which is add	What are the reasons for abandoning your	Does discount/free shipping i
6	12-12-2025 10:50:37	THARIKA THAHAS	18 – 24 years	Female	Sometimes	Amazon, Flipkart, Myntra	Yes	No	Product price too high	Yes
7	12-12-2025 10:54:55	Sadhana.S	18 – 24 years	Male	Sometimes	Amazon, Flipkart, Myntra, Meesho	Yes	Sometimes	Product price too high, Extra delivery charg	Yes
8	12-12-2025 11:06:12	Vani shree	18 – 24 years	Female	Frequently	Amazon, Flipkart, Myntra, Nykaa	Yes	Sometimes	Just browsing, Changed mind	Yes
9	12-12-2025 20:31:19	Indhu sri	18 – 24 years	Female	Sometimes	Amazon	Yes	No	Delivery time too long	Sometimes
10	12-12-2025 20:32:33	Rameshkumar	25 – 34 years	Male	Frequently	Amazon, Flipkart, Myntra, Meesho, Ajio	Yes	Yes		
11	12-12-2025 20:40:02	Kavin S	18 – 24 years	Male	Frequently	Amazon, Flipkart, Ajio	Yes	No	Extra delivery charges, Checkout takes too	Yes
12	12-12-2025 20:41:13	Anish Fathima Z	18 – 24 years	Female	Frequently	Amazon, Flipkart, Meesho, Nykaa	Yes	No	Just browsing, Delivery time too long	Yes
13	12-12-2025 20:42:39	Akshara S	18 – 24 years	Female	Sometimes	Amazon, Flipkart, Meesho	Yes	No	Extra delivery charges, Delivery time too lo	Yes
14	12-12-2025 20:45:36	Shalini T	18 – 24 years	Female	Frequently	Amazon, Flipkart, Ajio, Nykaa	Yes	No	Product price too high, Extra delivery charg	Yes
15	12-12-2025 20:46:05	Dharaneesh S	18 – 24 years	Male	Frequently	Flipkart, Myntra, Ajio	Yes	No	Product price too high, Delivery time too lo	Yes
16	12-12-2025 20:48:44	Saravanan S	45+ years	Male	Rarely	Flipkart	Yes	No	Just browsing	Sometimes
17	12-12-2025 20:50:38	Subha Devi S	35 – 44 years	Female	Sometimes	Amazon, Flipkart, Myntra, Meesho, Ajio, N	Yes	No	Checkout takes too long, Delivery time too	Yes
18	12-12-2025 20:52:31	SS	18 – 24 years	Male	Sometimes	Amazon	Yes	Yes		
19	12-12-2025 20:54:31	Lavanya thangavel	18 – 24 years	Female	Frequently	Amazon, Flipkart, Meesho	Yes	No	Changed mind	Yes
20	12-12-2025 21:05:21	S.Harisudhan	18 – 24 years	Male	Sometimes	Amazon, Flipkart	Yes	No	Delivery time too long	No
21	12-12-2025 21:13:35	Sivaharitha S	18 – 24 years	Female	Sometimes	Amazon, Flipkart, Myntra, Meesho	Yes	No	Extra delivery charges, Just browsing, Char	Yes
22	12-12-2025 21:14:03	Andrea	18 – 24 years	Female	Sometimes	Amazon, Myntra, Nykaa	Yes	Yes		
23	12-12-2025 21:20:25	Adharsini	18 – 24 years	Female	Rarely	Amazon, Flipkart	Yes	Yes		
24	12-12-2025 21:21:11	Dharun MR	18 – 24 years	Male	Rarely	Amazon, Flipkart	Yes	No	Product price too high, Just browsing, Deli	Yes
25	12-12-2025 21:41:47	KS	18 – 24 years	Female	Sometimes	Amazon	Yes	Sometimes	Extra delivery charges, Changed mind	Sometimes
26	12-12-2025 21:56:02	Siva	18 – 24 years	Male	Rarely	Amazon	Yes	Sometimes	Product price too high, Extra delivery charg	Yes

Form responses 1

Cleaned Sample Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Timestamp	Name	Age_Group	Gender	e_Shopping_Freq	eferred_Shopping_A	Added_To_Car	urchase_Statu	ndonment_Rea	Discount_Influence	kout_Impor	ter_Notificati	Suggestions_1	Suggestions_2	Reason_Category			
2	##################	Rakshith	18–24 years	Male	Sometimes	Myntra	Yes	No	product price t	Yes	Yes	Yes	improve the p	No suggestion	Price Issue			
3	##################	Rithikkesh B	18–24 years	Male	Frequently	Amazon, Flipkart, M	Yes	No	product price t	Yes	Yes	Yes	Needs to imp	No suggestion	Price Issue			
4	##################	Sangamithra S	18–24 years	Female	Frequently	Flipkart, Ajio	Yes	No	just browsing, c	Yes	Yes	Yes	No suggestion	No suggestion	Browsing Behavior			
5	##################	Subhagini	18–24 years	Female	Sometimes	Amazon, Flipkart	Yes	Sometimes	checkout takes	Sometimes	Yes	Sometimes	Not Answered	Not Answered	Delivery Time Issue			
6	##################	THARIKA THAHASIN	18–24 years	Female	Sometimes	Amazon, Flipkart, M	Yes	No	product price t	Yes	Yes	Yes	No suggestion	No suggestion	Price Issue			
7	##################	Sadhana.S	18–24 years	Male	Sometimes	Amazon, Flipkart, M	Yes	Sometimes	product price t	Yes	Yes	No	Honest produ	Not Answered	Price Issue			
8	##################	Vani shree	18–24 years	Female	Frequently	Amazon, Flipkart, M	Yes	Sometimes	just browsing, c	Yes	Yes	Sometimes	Provide perso	Provide persor	Browsing Behavior			
9	##################	Indhu sri	18–24 years	Female	Sometimes	Amazon	Yes	No	delivery time to	Sometimes	Yes	Sometimes	Improve prod	Improve produ	Delivery Time Issue			
10	##################	Ramashkumar	25–34 years	Male	Frequently	Amazon, Flipkart, M	Yes	Yes	not applicable	Not Answered	Not Answer	Not Answere	No suggestion	Nil	Technical Issue			
11	##################	Kavin S	18–24 years	Male	Frequently	Amazon, Flipkart, A	Yes	No	extra delivery c	Yes	Yes	Yes	It would be go	It would be goc	Delivery Cost Issue			
12	##################	Anish Fathima Z	18–24 years	Female	Frequently	Amazon, Flipkart, M	Yes	No	just browsing, c	Yes	Yes	Yes	Not Answered	Not Answered	Delivery Time Issue			
13	##################	Akshara S	18–24 years	Female	Sometimes	Amazon, Flipkart, M	Yes	No	extra delivery c	Yes	Yes	Yes	It would be be	It would be bet	Delivery Cost Issue			
14	##################	Shalini T	18–24 years	Female	Frequently	Amazon, Flipkart, A	Yes	No	product price t	Yes	Yes	Yes	Reduce the de	Not Answered	Price Issue			
15	##################	Dharaneesh S	18–24 years	Male	Frequently	Flipkart, Myntra, Aj	Yes	No	product price t	Yes	Yes	Sometimes	Flexible deliv	Product review	Price Issue			
16	##################	Saravanan S	45+ years	Male	Rarely	Flipkart	Yes	No	just browsing	Sometimes	Yes	Yes	I don't know	I don't have an	Browsing Behavior			
17	##################	Subha Devi S	35–44 years	Female	Sometimes	Amazon, Flipkart, M	Yes	No	checkout takes	Yes	Yes	Yes	It would be ea	It would be bet	Delivery Time Issue			
18	##################	SS	18–24 years	Male	Sometimes	Amazon	Yes	Yes	not applicable	Not Answered	Not Answer	Not Answere	No suggestion	Simple but luxu	Technical Issue			
19	##################	Lavanya thangave	18–24 years	Female	Frequently	Amazon, Flipkart, M	Yes	No	changed mind	Yes	Yes	Sometimes	Do some offer	You want to cre	User Decision			
20	##################	S.Harisudhan	18–24 years	Male	Sometimes	Amazon, Flipkart	Yes	No	delivery time to	No	Yes	Yes	Use More Cus	–Improve Web	Delivery Time Issue			
21	##################	Sivaharitha S	18–24 years	Female	Sometimes	Amazon, Flipkart, M	Yes	No	extra delivery c	Yes	Yes	No	Not Answered	Faster search	Delivery Cost Issue			
22	##################	Andrea	18–24 years	Female	Sometimes	Amazon, Myntra, N	Yes	Yes	not applicable	Not Answered	Not Answer	Not Answere	No suggestion	To have a close	Technical Issue			
23	##################	Adharsini	18–24 years	Female	Rarely	Amazon, Flipkart	Yes	Yes	not applicable	Not Answered	Not Answer	Not Answere	No suggestion	No	Technical Issue			
24	##################	Dharun MR	18–24 years	Male	Rarely	Amazon, Flipkart	Yes	No	product price t	Yes	Yes	No	Nil	Community for	Price Issue			
25	##################	KS	18–24 years	Female	Sometimes	Amazon	Yes	Sometimes	extra delivery c	Sometimes	Yes	Yes	Maybe we car	We can provid	Delivery Cost Issue			
26	##################	Siva	18–24 years	Male	Rarely	Amazon	Yes	Sometimes	product price t	Yes	Yes	No	Nope	Low price	Price Issue			
27	##################	Pavithra	18–24 years	Female	Rarely	Amazon, Flipkart, M	Yes	Sometimes	product price t	Yes	Yes	Sometimes	Look out for g	Purchase from	Price Issue			
28	##################	Gayathri Muthusa	18–24 years	Female	Sometimes	Flipkart, Meesho	Yes	Sometimes	delivery time to	Sometimes	Yes	Sometimes	We have to sh	We have to shi	Delivery Time Issue			
29	##################	Abdul Muksith	18–24 years	Male	Sometimes	Amazon, Flipkart, M	Yes	Yes	not applicable	Not Answered	Not Answer	Not Answere	No suggestion	I think nowada	Technical Issue			

Source Code

Data Preprocessing Code

```
 import pandas as pd
import numpy as np
# Load dataset
df = pd.read_excel("dataset.xlsx")
# Remove duplicates
df = df.drop_duplicates()
# Handle missing values
df = df.fillna(method='ffill')
# Standardize text data
df.columns = df.columns.str.strip()
df = df.applymap(lambda x: x.strip() if isinstance(x, str) else x)
# Display cleaned dataset
print(df.head())
```

Age Group

```
df["Age_Group"].unique()
df["Age_Group"] = df["Age_Group"].replace({
    "18 - 24 years": "18 – 24 years"
})
df["Age_Group"].value_counts()
```

Gender Distribution

```
df["Gender"].value_counts()
# Online_Shopping_Frequency
df["Online_Shopping_Frequency"].value_counts()
# Purchase_Status
df["Purchase_Status"].value_counts()
```

```
# Abandonment_Reason
df["Abandonment_Reason"].value_counts()
df["Abandonment_Reason"] = df["Abandonment_Reason"].str.lower()
# Split by comma
reasons_split = df["Abandonment_Reason"].str.split(",")

# Convert list into rows
reasons_exploded = reasons_split.explode()
reasons_exploded = reasons_exploded.str.strip()
reasons_exploded = reasons_exploded[reasons_exploded != "not applicable"]
reason_counts = reasons_exploded.value_counts()
print(reason_counts)

#Age vs Purchase
pd.crosstab(df["Age_Group"], df["Purchase_Status"])
#Gender vs Purchase
pd.crosstab(df["Gender"], df["Purchase_Status"])
#Purchase vs Abandonment
df["Purchase_Status"].value_counts()

#percentage
purchase_counts = df["Purchase_Status"].value_counts()
purchase_percentage = (purchase_counts / purchase_counts.sum()) * 100
print(purchase_percentage.round(2))

# Cart Abandonment Rate
abandon_rate = purchase_percentage.get("No", 0)
print("Cart Abandonment Rate:", round(abandon_rate, 2), "%")
```

```
#Reason category
def categorize_reason(reason):
    reason = str(reason).lower()

    if "price" in reason or "cost" in reason:
        return "Price Issue"
    elif "delivery" in reason or "shipping" in reason:
        return "Delivery Issue"
    elif "payment" in reason:
        return "Payment Issue"
    elif "checkout" in reason:
        return "Checkout Issue"
    elif "browse" in reason or "later" in reason:
        return "Browsing Behavior"
    else:
        return "Other"
df["Reason_Category"] = df["Abandonment_Reason"].apply(categorize_reason)
df["Reason_Category"].value_counts()

#app preferred
df["Preferred_Shopping_App"].value_counts().head(10)
#app count
df["Preferred_Shopping_App"] = df["Preferred_Shopping_App"].str.lower()
apps_split = df["Preferred_Shopping_App"].str.split(",")
apps_exploded = apps_split.explode()
apps_exploded = apps_exploded.str.strip()
app_counts = apps_exploded.value_counts()
print(app_counts)
```

```
#age vs abondenment
age_abandon = pd.crosstab(df["Age_Group"], df["Purchase_Status"])
print(age_abandon)
# Convert to percentage
age_abandon_percent = age_abandon.div(age_abandon.sum(axis=1), axis=0) * 100
print(age_abandon_percent.round(2))
# gender vs abondenment
gender_abandon = pd.crosstab(df["Gender"], df["Purchase_Status"])
print(gender_abandon)
# gender abandon percentage
gender_abandon_percent = gender_abandon.div(gender_abandon.sum(axis=1), axis=0) * 100
print(gender_abandon_percent.round(2))

#Risk score
# Function to calculate risk score
def calculate_risk(row):
    score = 0

 # Purchase behavior
    if row["Purchase_Status"] == "No":
        score += 3
    elif row["Purchase_Status"] == "Sometimes":
        score += 2

 # Shopping frequency
    if row["Online_Shopping_Frequency"] == "Rarely":
        score += 2
    elif row["Online_Shopping_Frequency"] == "Sometimes":
        score += 1
```

```
 # Abandonment reason (price/delivery related)
    reason = str(row["Abandonment_Reason"]).lower()
    if "price" in reason or "delivery" in reason:
        score += 1
    return score
df["Risk_Score"] = df.apply(calculate_risk, axis=1)
df[["Purchase_Status", "Online_Shopping_Frequency", "Risk_Score"]].head()

#risk level
def risk_level(score):
    if score >= 4:
        return "High Risk"
    elif score >= 2:
        return "Medium Risk"
    else:
        return "Low Risk"
df["Risk_Level"] = df["Risk_Score"].apply(risk_level)
df["Risk_Level"].value_counts()
# Assume average cart value
avg_cart_value = 1000
# Count abandoned customers
abandoned_count = df[df["Purchase_Status"] == "No"].shape[0]
# Total revenue loss
total_loss = abandoned_count * avg_cart_value
print("Total Abandoned Customers:", abandoned_count)
print("Estimated Total Revenue Loss: ₹", total_loss)
# loss per reason
loss_per_reason = reason_counts * avg_cart_value
print(loss_per_reason.head(10))
```

```
#top 3 loss
top3_loss = loss_per_reason.head(3)
print("Top 3 Reasons Causing Maximum Revenue Loss:")
print(top3_loss)

# Possible improvement scenarios
reduction_rates = [0.10, 0.20, 0.30]

for rate in reduction_rates:
    saved_customers = int(abandoned_count * rate)
    saved_revenue = saved_customers * avg_cart_value
    print(f"If abandonment reduces by {int(rate*100)}%:")
    print("Recovered Customers:", saved_customers)
    print("Revenue Saved: ₹", saved_revenue)
    print("------")
```

Final Dashboard